Design of 16x16 Bit Comparator

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Abstract

We present in this paper the design and implementation of a CMOS-based 16-bit magnitude comparator. First, a simple 1-bit comparator is designed, and it is then scaled up to a 4-bit comparator and finally expanded to a 16-bit comparator. Optimizing the comparator's speed, power consumption, and area efficiency is the main goal of this project. To achieve low power consumption without sacrificing performance, a VDD of 0.8V is used. A variety of logic gates, such as NOR, XNOR, inverter, AND, and OR gates, are incorporated into the design. These gates are carefully sized to balance the performance of PMOS and NMOS transistors. The simulation results show that the designed comparator meets the performance metrics that were intended, which qualifies it for low-power and high-speed applications in digital systems.

Introduction

The development of effective comparator designs, which are essential parts of many digital systems, has been fueled by the need for high-speed digital circuits. Comparators are used to compare binary numbers and ascertain their relative magnitudes in a variety of applications, including arithmetic operations, signal processing, and data sorting. In this work, we concentrate on developing a 16-bit magnitude comparator with CMOS technology, which provides a good trade-off between speed and power consumption. The first step in the design process is to create a 1-bit comparator, which compares two binary single-bit numbers. This is the most basic type of comparator. The 1-bit comparator is then instantiated four times to expand this basic unit into a 4-bit comparator. Ultimately, a 16-bit comparator is produced by making four copies of the 4-bit comparator. The VDD of 0.8V is used in the design as a deliberate decision to reduce power consumption and maintain sufficient speed. To achieve the desired logic functions, we use a combination of basic logic gates in our design, including NOR, XNOR, inverter, AND, and OR gates. To guarantee peak performance, the gates are carefully sized, with special attention to how to balance the PMOS and NMOS transistors. The product is a low-power, high-speed comparator that can meet the demanding specifications of contemporary digital systems while also being space-efficient.

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2.Comparator

A magnitude digital comparer is combinational circuit that compares two binary numbers to determine their relationship. It has two inputs terminals, one for each binary number (A and B), and three output terminals indicating the comparison results: one for A >B, one for A=B, and one for A < B

Figure 1:N-bit Comparator

2.1 1-Bit Comparator

A comparator used to compare two bits is called a single-bit comparator.
It consists of two inputs each for two single-bit numbers and three outputs to generate less than, equal to and greater than between two binary numbers.

A	B	A<B	A=B	A>B
0	0	0	1	0
0	1	1	0	0
1	0	0	0	1
1	1	0	1	0

Figure 2:Truth Table for 1 bit comparator

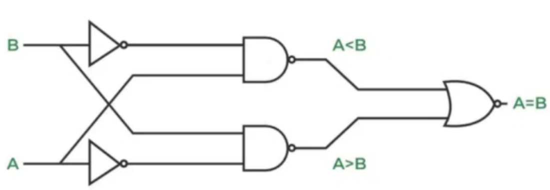

Figure 3:Block Diagram for 1bit comparator

2.2 4-Bit Magnitude Comparator

A 4-bit magnitude comparator. It consists of eight inputs each for two four-bit numbers and three outputs to generate less than, equal to, and greater than between two binary numbers.

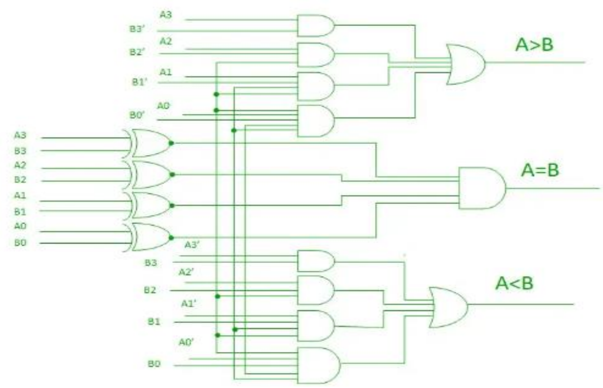

Figure 4:4 Bit comparator

the condition of A>B:

1. If A3 = 1 and B3 = 0
2. If A3 = B3 and A2 = 1 and B2 = 0
3. If A3 = B3, A2 = B2 and A1 = 1 and B1 = 0
4. If A3 = B3, A2 = B2, A1 = B1 and A0 = 1 and B0 = 0

for A<B:

1. If A3 = 0 and B3 = 1
2. If A3 = B3 and A2 = 0 and B2 = 1
3. If A3 = B3, A2 = B2 and A1 = 0 and B1 = 1
4. If A3 = B3, A2 =B2, A1 = B1 and A0 = 0and B0= 1

The condition of A=B is possible only when all the individual bits of one number exactly coincide with the corresponding bits of another number.

A	B	C	D	g	l	e
0	0	0	0	0	0	1
0	0	0	1	0	1	0
0	0	1	0	0	1	0
0	0	1	1	0	1	0
0	1	0	0	1	0	0
0	1	0	1	0	0	1
0	1	1	0	0	1	0
0	1	1	1	0	1	0
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	0	0	1
1	0	1	1	0	1	0
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	0	0	1

Figure 5: 4 bit comparator truth table

3. Schematic and Layout

3.1 Design of 1x1- bit comparator

A 1-bit comparator is a device which compare two 1- bit binary numbers and provides output.

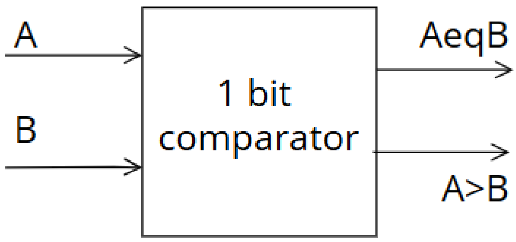

Figure 6:1 bit comparator

in our project we build it from basic gates, 2 Inverter gate, nor and xnor gates. It takes two inputs A and B and provides 2 output equal and greater than as shown in the figure 7 below which represents the schematic for 1 bit comparator.

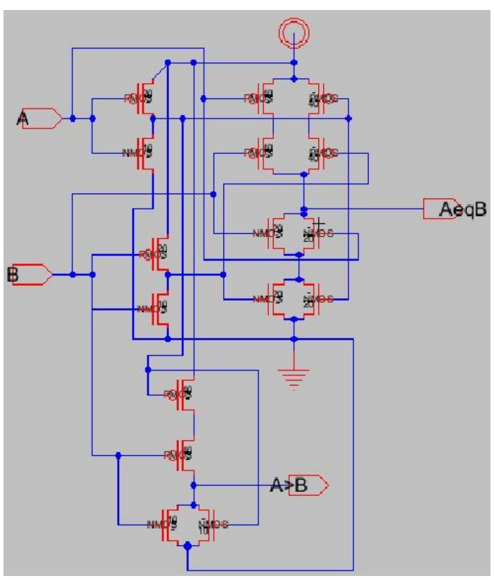

Figure 7:1 bit comparator schematic

The figure 8 shows the layout for 1 bit comparator.

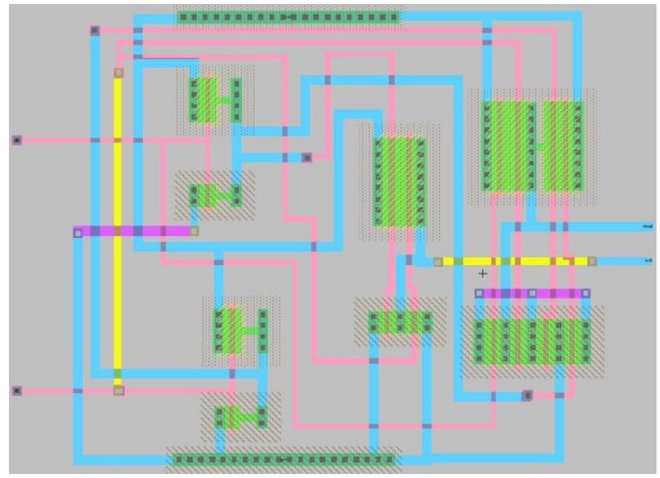

Figure 8: 1 bit comparator layout

3.2 Design of 4x4-bit comparator

A 4-bit comparator is a device which compare two 4- bit binary numbers and provides output

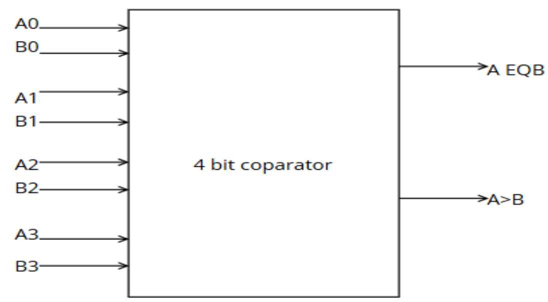

Figure 9: 4 bit comparator

We build it from four 1 bit comparator and from 2,3,4 input AND gates and 4 input OR gate. It takes also two inputs (4 bits) and provide 2 outputs equal and greater than as shown in figure 10 below which represents the schematic for 4 bit comparator.

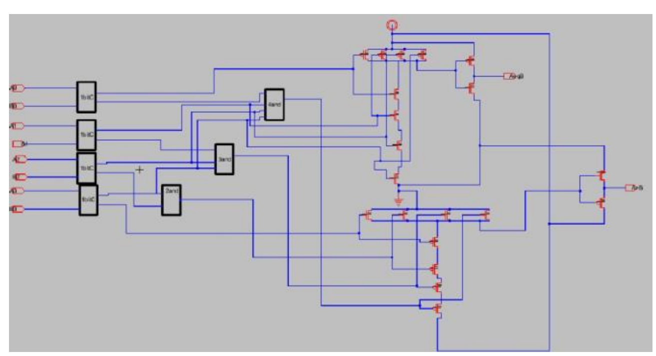

Figure 10: 4 bit schematic

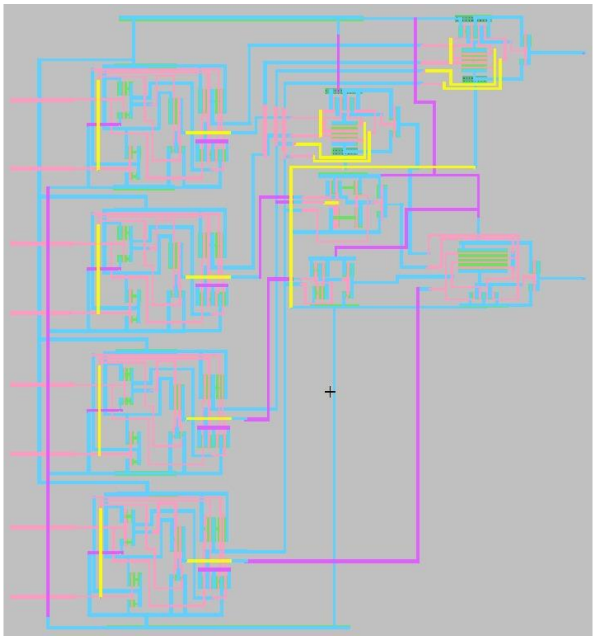

Figure 11: 4-bit layout

3.3 Design of 16x16-bit comparator

A 1-bit comparator is a device which compare two 16- bit binary numbers and provides output.

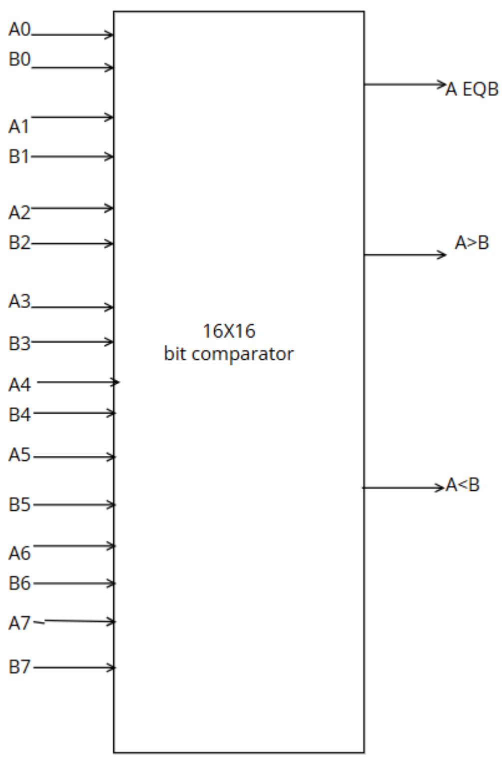

Figure 12:16 bit comparator

We build the 16x16 bit comparator from four 4 bit comparator and basic gates 2,3,4 input AND gate and 4 input OR gate. Additionally we add an nor gate to provide the less than output. The circuit takes two inputs each input represents from 8 bit, and it provide three outputs equal, less than, greater than as shown in the figure 13 which represents the schematic of 16 bit comparator.

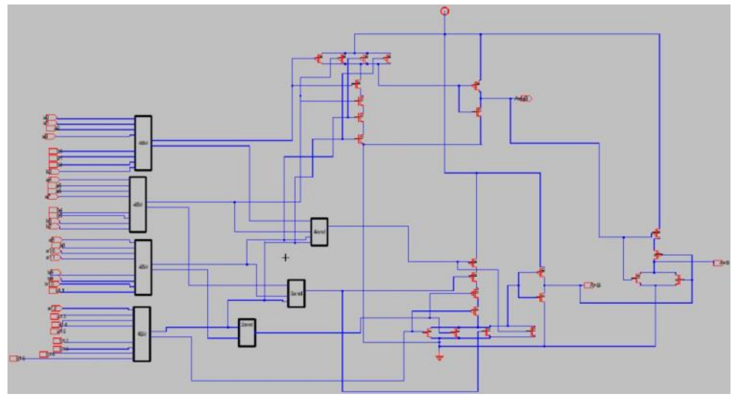

Figure 13: 16 bit schematic

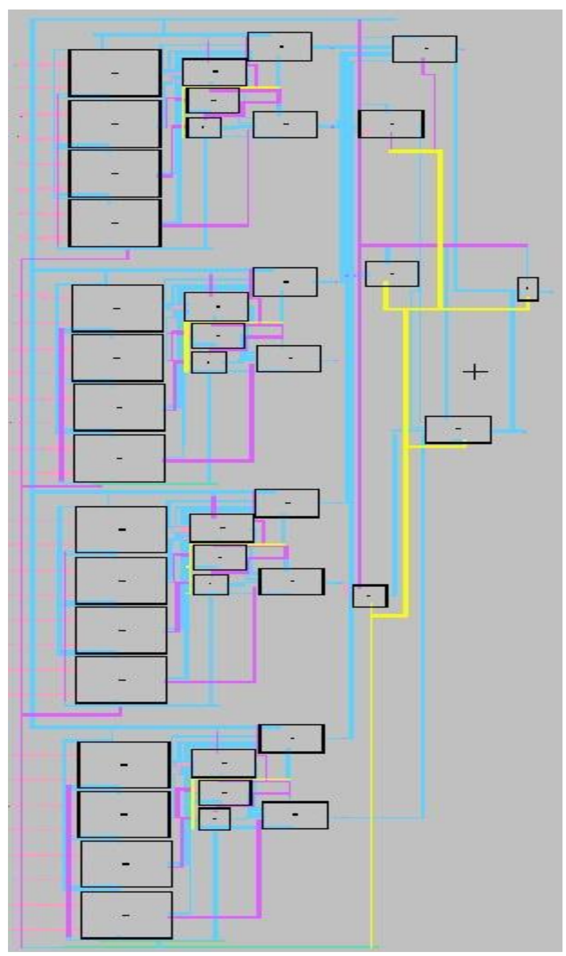

Figure 14: 16 bit layout

4.Simulation Results

4.1 1x1 bit comparator

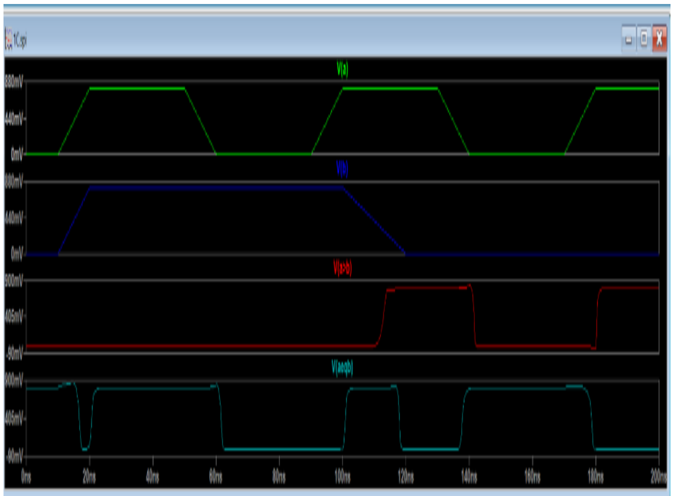

Figure 15: 1x1 bit schematic simulation

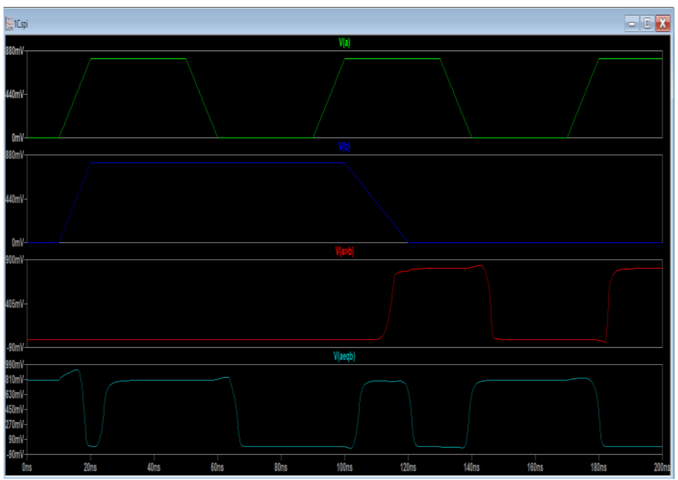

Figure 16: 1x1 bit layout simulation

The 1-bit comparator's simulation results show that the circuit accurately carries out its intended function, correctly indicating when one input is greater than or equal to the other through the outputs V(a>b) and V(a=b). The binary values represented by the input signals V(a) and V(b) transition between 0V and 0.8V, and the output signals react to these changes accordingly. In particular, When V(a) is greater than V(b), V(a>b) rises, and When both inputs are equal, V(a=b) rises. These transitions' timing corresponds with the input changes, which shows that the comparator functions with little delay—a critical characteristic for high-speed applications. Overall, the simulation validates that the design of the 1-bit comparator is correct, offering a solid basis for scaling up to more intricate multi-bit comparators such as the 16-bit

4.2 4x4 bit comparator

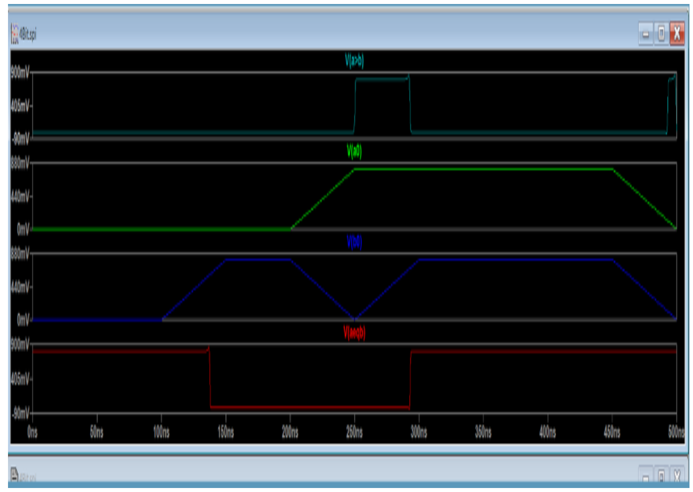

Figure 17: 4x4 bit schematic simulation

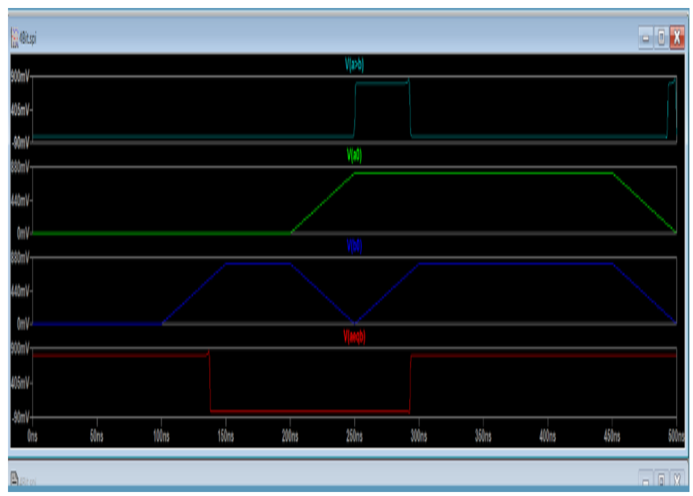

Figure18: 4x4 bit layout simulation

The 4-bit comparator simulation shows that the circuit can distinguish between equal and greater-than conditions for the input signals with accuracy. For the 4-bit numbers under comparison, the least significant bits are represented by the inputs V(a0) and V(b0). Between 250 and 300 ns, the output V(a=b) (cyan) rises, showing that the two 4-bit numbers are equal at this time. In the meantime, after 300 ns, V(a>b) (red) transitions high, indicating that the 4-bit number represented by V(a) has increased in value relative to V(b). These output signals' timing perfectly matches the input changes, demonstrating the comparator's proper operation. The 4-bit comparator design is validated and its scalability to higher-bit comparisons is supported by this accurate performance under all conditions.

4.3 16x16 bit comparator

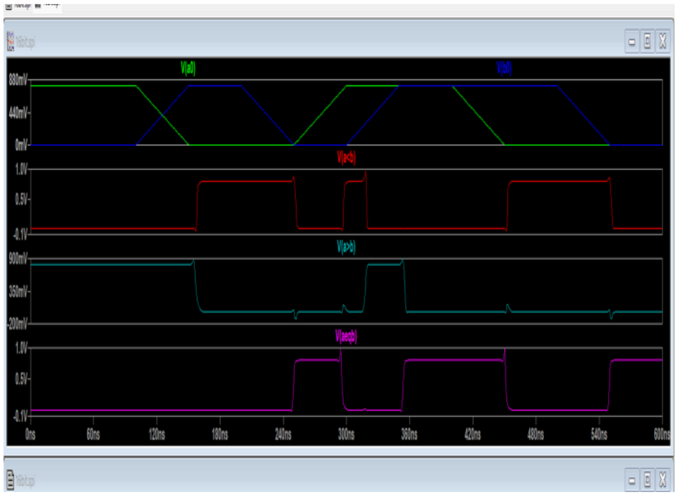

Figure 19: 16x16 bit schematic simulation

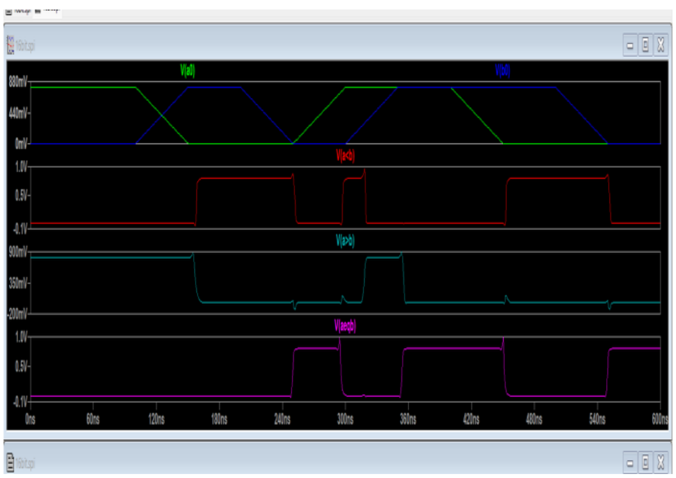

Figure 20: 16x16 bit layout simulation

These output signals transition in a way that makes sense given the variations in the input. The outputs promptly reflect the input conditions, indicating the accuracy and speed of the 16-bit comparator design. The timing of the transitions is consistent. But the simulation also shows that there are a few minor glitches, which are especially apparent in the output signals. When the inputs are changing states during switching events, there are brief, inadvertent transitions known as glitches. They are probably caused by brief discrepancies in the propagation delays of various circuit paths, which can lead to an incorrect comparison being temporarily registered at the output. Even with these shortcomings, the comparator eventually settles to the correct states such that V(a<b), V(a>b), and V(aeqb) accurately resolve the intended comparison results once the inputs stabilize. Even though the errors don't have a big effect on the output, their existence indicates that more optimization might be helpful to improve the circuit's dependability, especially in settings where noise sensitivity is an issue. All in all, these findings verify that the circuit is functioning as planned, efficiently comparing 16-bit binary numbers and delivering dependable outputs for conditions that are greater than, less than, and equal. They also point out possible areas for improvement to guarantee even more robustness in high-speed applications.

5. 16x16 bit comparator power and speed and area

For the sizing we found that the best size for PMOS is 20 and for NMOS 10, where at these sizes the rising and falling time close to each other.

Power = Imax * VDD, we obtained the result by code.

```
pavg: AVG(i(vdd)*v(vdd))=-5.76187e-005 FROM 0 TO 5e-007
td=5.26757e-008 FROM 1.59375e-007 TO 2.12051e-007
vaeqb=2.07982e-009 FROM 2.89147e-007 TO 2.91226e-007
vaeqbt=3.79978e-009 FROM 2.07065e-007 TO 2.10865e-007
vagb=3.38206e-009 FROM 3.97324e-007 TO 4.00706e-007
vagbt=6.18448e-009 FROM 2.93028e-007 TO 2.99213e-007
vltb=2.22294e-009 FROM 2.10625e-007 TO 2.12848e-007
vltbt=1.81857e-009 FROM 2.91811e-007 TO 2.93629e-007
```

Figure21: Power and Delay results

The comparator's timing metrics and power analysis highlight a number of important performance factors. $5.76187×10^{-5}$ W is the average power consumption, which indicates that the circuit operates with relatively low power, making it appropriate for energy-efficient applications. The time it takes for signals to travel through the circuit is reflected in the transmission delay (td), which is 5.267577×10^{-8} s. This value indicates how quickly the comparator reacts to changes in input. The signals vaeQB, vaeQBT, vagb, and vltBT have rise and fall times ranging from 2.07982×10^{-9} s to 6.18448×10^{-9} s, suggesting that the circuit switches between logical states effectively and with little delay. These numbers attest to the comparator's speed, power efficiency, and short reaction times, which qualify it for use in high-speed digital systems.

The speed is determined by calculating the propagation delay for each output, equal greater than, less than.

```
prop_delay=-1.18445e-007 FROM 2.25e-007 TO 1.06555e-007
propa_delay=1.27995e-007 FROM 2.25e-007 TO 3.52995e-007
propb_delay=-7.76957e-008 FROM 2.25e-007 TO 1.47304e-007
```

Figure22: propagation delay for each output

The propagation delay results show that the comparator circuit's various paths have different speeds. The negative delay values, like −7.76957×10 ~^−8 and −1.18445×10 −^7}, point to possible problems with timing measurements or show that the output signal transitions before the input signal, which could be caused by setup anomalies or simulation artifacts. Conversely, however, the positive delay value The actual time it takes for a signal to travel through the comparator is reflected in 1.27995×10^−7 s. The existence of both positive and negative delays suggests irregularities that may affect the circuit's overall speed and dependability.

Area

SIZE: 1762.5 x 4364.5

6. Conclusion

In this paper, we have successfully used CMOS technology to design and implement a 16-bit magnitude comparator. The design process shows the viability and efficiency of applying hierarchical design techniques in complex digital circuits. It started with a basic 1-bit comparator and was methodically scaled to a 16-bit comparator. We were able to strike a balance between low power consumption and high performance by carefully choosing a 0.8V supply voltage, which qualifies the comparator for use in contemporary digital systems where efficiency and speed are vital. NOR, XNOR, inverter, AND, and OR gates, among other carefully sized logic gates, are included in our design to guarantee that the comparator performs well within the intended parameters. The results of the simulation verified that the comparator, with its optimized power consumption, speed, and area utilization, satisfies the necessary performance metrics. The work showcased here emphasizes how crucial careful component sizing and design are to building high-performance digital circuits. This project's 16-bit comparator is a dependable solution for high-speed, low-power operations, and it can be used in a wide range of digital applications. Additionally, this study offers a strong basis for further research, such as investigating additional optimizations and modifications for various technological nodes or application-specific needs.

7. References

https://www.geeksforgeeks.org/magnitude comparator-in-digital-logic/

https://i0.wp.com/study.com/cimages/mult images/16/turth_table801856356999066057.png?strip=all

https://online.visual-paradigm.com/diagrams/features/logic-diagram-software/